

COMP 1111 1. Midterm

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I. QUESTION (25 POINTS)

There are four cats and three types of food for these cats in our house. Not every cat eats every type of food. First cat "Zeytin" eats only second type of food. Second cat "Pamuk" eats first and third type of food. Third cat "Duman" eats second and third type of food. Fourth cat "Şaban" eats all type of foods. Write a program which takes the index of the cat and index of the food and displays "Yes" on the screen if the given cat eats the given type of food, "No" otherwise". Use switch structure for this question.

```
Enter cat index: 2
Enter food index: 1
Yes
```

```
Enter cat index: 3
Enter food index: 1
No
```

II. QUESTION (25 POINTS)

Write a program which will read a four digit integer and finds out if the number is divisible by 11 according to the following algorithm: Let the four digit number be "abcd". You need to calculate the odd digit sum as $b + d$, and the even digit sum as $a + c$. If the difference between the odd and even digit sums is divisible by 11, then the four digit number is divisible by 11. Otherwise, the number is not divisible by 11.

```
Enter number: 1331
Divisible
```

```
Enter number: 1321
Not divisible
```

III. QUESTION (25 POINTS)

Write a program which takes two integers a and b from the user. The program will print the number of integers which are divisible by 3 between [a, b]. Loops are not allowed in this problem.

```
a: 5
b: 126
Numbers divisible by 3: 41
```

IV. QUESTION (25 POINTS)

Write a program which simulates the trip of a car. The car starts from an initial point and travels until the first traffic light. If the first light is green, the car continues, otherwise it waits for the green signal. Afterwards, the car continues and travels until the second traffic light. Again, If the second light is green, the car continues, otherwise it waits for the green signal. The traffic lights are green at time 0. The traffic lights are green or red for the same time interval. The user enters (i) the speed of the car in m/s, (ii) the length of the road (in meters) from initial point to the first light, (iii) time interval for the first light (in seconds), (iv) the length of the road (in meters) from the first light to the second light, (v) time interval for the second light (in seconds). The program will output the time required to finish the trip.

```
Speed: 4
1. Road Length: 100
1. Time Interval: 20
2. Road Length: 100
2. Time Interval: 20
Simulation Time: 80
```